ChargeGrid AI

A planning tool that tells you **where to build EV charging stations**, **how big to make them** and **whether they will make money**. Built first for Pune and the Mumbai–Pune highway.

What it does, in one paragraph

India is adding electric vehicles fast, but chargers are often in the wrong places: crowded in some areas, missing in others. ChargeGrid AI looks at a city on a map, works out how many EVs there are and will be, finds the areas where drivers will not find a charger, and suggests the best places to build new ones within a given budget. For each suggested site it also works out how many chargers are needed, what it will cost, when it pays back, whether solar panels and batteries help, and whether the local electricity grid can handle it. You can also just ask it a question in plain English and it does the planning for you.

Who it is for

- **Charging companies** (CPOs) deciding where to put their next stations.
- **City and state planners** who want fair coverage across neighbourhoods.
- **Electricity companies** (DISCOMs) checking where new load will hit the grid.
- **Investors and consultants** who need costs, returns and risks for each site.

How it works: the 6 steps

Step	In simple words
1. Collect data	Pulls in maps (roads, shops, malls, petrol pumps), the chargers that already exist, population and vehicle registrations, and places each one on a grid of small hexagon cells.
2. Predict demand	Forecasts how many EVs there will be each year (slow, normal and fast growth) and how much public charging they will need. The **gap** is demand that existing chargers cannot cover.
3. Find candidate sites	Lists possible places to build: malls, fuel stations, highway spots, gap areas or your own sites. Removes duplicates and sites that break rules (too close to another charger, too far from power, and so on), and says why each one was rejected.
4. Score sites	Rates every site on 8 things (demand nearby, traffic, competition, grid, land and more) using the area a driver can reach in 10 minutes. Shows how stable the ranking is.
5. Make a plan	Given a budget and a site limit, picks the best set of sites and how many chargers each gets, to cover as much demand as possible. It can explain why a good site was left out and redo the plan for what-ifs in a few seconds.
6. Check each site	For every chosen site: number of chargers needed so people do not wait, costs and 10-year profit (with best and worst cases), solar + battery options, and the effect on the local grid.

The screens

The top menu has: **Workspace · Area planner · Plans · Scenarios · Assistant · Data · Admin**. When the app is running, every screen is at **http://localhost:5173** plus the address shown.

Screen (address)	What you see	What you can do
Home **/**	Welcome page with a short introduction.	Jump to the main screens.
Workspace **/workspace**	Big map of the region with existing chargers and coloured hexagons for EV demand and the charging gap. Side panel with charger counts.	Switch map layers, click a charger to see where its data came from, switch to a Google basemap.
Area planner **/areas/…**	One area (for example, Pune): EV count forecast chart, EVs by type (two-wheeler, car, bus…), and existing chargers by power, connector and operator.	Compare slow, normal and fast EV growth for that area.
Plans **/plans**	List of planning runs (scenarios) you have made.	Start a new plan: pick the region, budget, site limit and score weights.
Plan result **/runs/…**	Results of one plan, in tabs: **Funnel** (how many sites at each step), **Ranked sites**, **Plan** (chosen sites on the map), **Strategies**, **Phasing**.	Ask "why not this site?", try what-ifs (bigger budget, faster adoption), compare strategies (max coverage, fairness, lowest cost, most profit), plan in phases for 2026/2028/2030, export to Excel, PDF or GeoJSON.
Site details **(inside a plan)**	For each chosen site: **Sizing**, **Finance** and **Energy** tabs.	See the number and type of chargers, costs, profit and payback, and the solar + battery recommendation.
Grid impact **/runs/…/grid**	24-hour electricity load the plan adds, split by vehicle type or charger type, and where it connects to the grid.	See which substations or transformers get busy and what upgrades might cost.
Google site view **/runs/…/sites/…/google**	A site on Google Maps with Google Solar data for the nearest roof.	See roof solar potential and sized options for that roof.
Scenarios **/twin**	Region-wide "what if" view with demand and gap maps.	Try ready scenarios: add 10,000 fast chargers, EV adoption +30%, where the grid limits growth.
Assistant **/assistant**	A chat screen with an AI planner (Claude).	Ask questions or request a plan in plain English. It shows its steps, asks you to confirm before planning, and gives a report where every number is checked.
Data catalogue **/data**	Every data source used, how fresh and how trustworthy it is, and a **readiness score** for each region.	See what data is real, what is a placeholder, and what to fix next.

Screen (address)	What you see	What you can do
Admin **/admin**	Members, settings and the audit log (who did what).	Manage users and roles, change settings and roll them back. Admins only. (Sign-in is at /login, by email or company single sign-on.)

A typical journey

- Open **Workspace** to see where chargers are today and where the gaps are.
- Open **Area planner** to see how fast EVs will grow in that area.
- Go to **Plans**, create a plan with a budget (for example, Rs 50 crore and up to 20 sites).
- Open the result: check the chosen sites, try a what-if, compare strategies.
- Open a site to see charger count, cost, payback and solar options; open **Grid impact** for the grid.
- Export the plan to Excel or PDF to share. Or do all of this by asking the **Assistant**.

What is real and what is a placeholder

Important: the app works end to end, but some inputs are stand-ins until real data is added. The app labels these on screen, and every suggested site still needs a visit on the ground.

Real data loaded	Still placeholders or missing
OpenStreetMap roads, places and points of interest	Vehicle registrations (VAHAN): synthetic stand-in
Existing chargers from OSM and Open Charge Map	Costs, tariffs and charger prices: illustrative
Population (WorldPop 2025, 1 km grid)	Some rule thresholds and demand settings
Solar data (NASA POWER, 2024)	Electricity company (DISCOM) grid data: none yet
Government policy documents for the assistant	Live traffic counts, income and housing data

Regions

Ready: Pune Metropolitan Region and the Mumbai–Pune corridor. **Registered for later:** Maharashtra and six other metros; the Data screen shows what each one still needs.

How to run it on this Mac

1. Start Docker Desktop.
2. In the ChargeGridAI folder run: `docker compose up -d --no-build`
3. Open **http://localhost:5173** in the browser.
4. To stop: `make down`

The AI Assistant also needs an Anthropic API key in the `.env` file.

Under the hood (short version)

Website: React. Server: Python (FastAPI). Database: PostgreSQL with map support (PostGIS). Drive-time maps: Valhalla. Optimisation: OR-Tools. AI agents: Claude. Everything runs in Docker.